THE APPARATUS

BY

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RESEARCH AND THEORY
THE CONSTANTS, THE UNIT, AND THE INSTRUMENT THAT DRAWS THE RESULT

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1

INFORMATION THEORY ON THE PLANE

This chapter arranges what the information-theoretic cryptanalysis literature says against what this work measures, and identifies the one axis that arrangement runs along.

Thirty-one hypotheses in this work are graded on collision entropy, which is Rényi order two. Order is a continuous axis running from zero to infinity, each point of it answering a different question about the same distribution, and the grading does not vary it. Order two is fixed by the instrument and not by the question: a collision counter measures it.

1.1 RESULTS FROM THE LITERATURE

1.1.1 *Guessing is governed by order one half, not order two*

Arikan's inequality [1] bounds the expected number of guesses for a user with the optimal guessing sequence:

$$E[G] \le \tfrac{1}{2}\Big[\sum_k \sqrt{p_k}\Big]^2 + \tfrac{1}{2} = \frac{2^{H_{1/2}(X)} + 1}{2}$$

with the Rényi entropy [12] of order α written $H_\alpha(X) = \frac{1}{1-\alpha}\log\sum_x P(x)^\alpha$, and with the property that carries the argument: H_α is strictly decreasing in α unless the distribution is uniform on its support.

Massey's earlier bound [10] is in Shannon entropy [13], and Malone and Sullivan [9] show why Shannon entropy is not a substitute for guesswork. Arikan's bound is sharper and names a *different order*: the quantity governing the cost of searching is $H_{1/2}$, whose defining sum is $\sum\sqrt{p}$. That sum is dominated by the many small probabilities and not by the few large ones. **Search cost is a tail statistic.**

1.1.2 *Collisions are governed by order two, and balance is that*

Bellare and Kohno [5] define the balance of a hash h with pre-image sizes d_i as

$$\mu(h) = \log_r\Big[\frac{d^2}{d_1^2 + \cdots + d_r^2}\Big]$$

and note immediately that $1/r^{\mu(h)}$ is the probability that $h(a) = h(b)$ for a, b drawn independently from the domain. Their Proposition 5.2 gives $0 \leq \mu(h) \leq 1$, zero exactly for a constant function and one exactly for a regular one, and Corollary 6.4 gives the birthday cost as $Q_h(c) \leq 1 + 2.36\sqrt{2c}\, r^{\mu(h)/2}$.

Balance is collision entropy rescaled: $\mu(h) = H_2 / \log_2 r$. It is the H_2 of this work in a different coordinate, and it answers the collision question, which is not the search question.

1.1.3 *Distinguishing cost is a divergence*

Baignères, Junod and Vaudenay [4, 3] put the shape of an optimal distinguisher on a hypothesis-testing footing: the optimal statistic is the log likelihood ratio, and the number of samples needed to separate P from uniform scales as the reciprocal of a divergence between them, with Kullback-Leibler [8] and Chernoff information as the exact exponent and squared Euclidean distance as the usable approximation. The chi-squared method [6] gives the crisp form, advantage over n samples bounded by $\sqrt{n\chi^2/2}$.

This is the principled replacement for the sigma counts this work reports. A sigma count says how surprising one statistic is. A divergence says how many samples any distinguisher whatsoever would need, the number deciding whether a structure is worth anything.

1.1.4 *Iterated functions lose entropy in a known amount*

Flajolet and Odlyzko [7] work through about twenty parameters of a uniformly random map on n points. The ones that bite here: the image is $(1 - 1/e)n$, leaving $1/e = 36.79\%$ of the range unreachable in one step; the giant component is $0.75788n$; rho and tail lengths go as $\sqrt{\pi n/8}$. Iteration entropy [2] follows the same loss through repeated application.

The holes are not a defect. **A random function is supposed to have them, and supposed to have exactly $1/e$ of them.**

1.2 THE ARRANGEMENT: ORDER IS THE AXIS

Every result above is a statement about the same object, the output distribution, read at a different Rényi order.

order	reads	what it answers	who
$\alpha = 0$	size of the support	how many values never occur, the holes	Flajolet-Odlyzko
$\alpha = 1/2$	$\sum \sqrt{p}$	cost of guessing, the search	Arikan
$\alpha = 1$	Shannon	information content	Massey
$\alpha = 2$	collision probability	cost of colliding, balance	Bellare-Kohno
$\alpha = \infty$	largest single probability	the biggest bump	min-entropy

Holes at one end, bumps at the other, guessing and collisions at two interior points. The plane is one plane; the order is the angle it is viewed from. The grading in this work views it from one angle.

1.2.1 *What a flat plane predicts along that axis*

Write the bin counts as $c_i = m(1 + e_i)$ with $m = d/r$ the exact mean. For a random function the e_i are small and the deficit $\log_2 r - H_\alpha$ expands to

$$\text{deficit}(\alpha) = \frac{\alpha \,\text{var}(e)}{2 \ln 2}, \qquad \text{var}(e) = \frac{r-1}{d} \text{ exactly}$$

The deficit profile of a random function is **a straight line through the origin in** α, which gives a test with no free parameter at all:

$$\text{deficit}(2)/\text{deficit}(1/2) = 4 \quad \text{exactly}$$

Every scale cancels out of a ratio between two orders: domain size, bin count, window position, how many samples were taken. Curvature in that line, or a ratio away from four, is structure that cannot be explained away by the size of the experiment.

Where the mean count falls to one the expansion stops applying and the counts go Poisson, whose moments about a unit mean are the Bell numbers $1, 2, 5, 15$. Both regimes occur here: a 32-bit window read over a 2^{32} domain is the Poisson case, and it is the only place a bin can be empty. That is where the holes live.

1.3 CONSEQUENCES FOR THE MEASUREMENTS

1.3.1 *Every advantage quoted from an H_2 measurement overstates the search advantage*

H_α is decreasing in α. Therefore $H_{1/2} \geq H_2$ and $\text{deficit}(1/2) \leq \text{deficit}(2)$, the first being about a quarter of the second for anything near flat. Every figure of the form *collision entropy is X bits below uniform, therefore there are X bits of advantage* is an answer to the collision question read as an answer to the search question, and it is the more generous of the two.

The concrete case: the rotational coherence hypothesis measures $H_2 = 16.503$ against a uniform 32, a deficit of 15.497 bits, with a permuted control at 32.069. That deficit is real and the control is sound. What it does not say is that 15.497 bits of search advantage exist, because search is governed by order one half. The supported statement is $\text{deficit}(1/2) \leq 15.497$, and the actual value needs the same histogram re-read at order one half. **That re-read is cheap and is not reported here.**

This is not a small correction of presentation. It is the difference between the quantity measured and the quantity the work is about.

1.3.2 *Bellare and Kohno's own SHA-1 numbers have an unsubtracted null*

Their full version computes the balance of SHA-1 restricted to a 2^{32} domain at byte-aligned output windows and reports the values are high. Subtracting the random-function null from their values:

window	their measured deficit	random function predicts	ratio
8 bits	1.0462×10^{-8}	1.0707×10^{-8}	0.9771
16 bits	1.3770×10^{-6}	1.3758×10^{-6}	1.0009
24 bits	2.34378×10^{-4}	2.34356×10^{-4}	1.0001

Their own resolution at width 8 is $\sqrt{2/255} = 8.8\%$, and 0.977 sits well inside it. **All of SHA-1's measured imbalance is the imbalance any function drawn at random would show, to four significant figures.** The paper says the balance is high. The sharper statement available from the same table is that the deficit is exactly generic, and that is a much stronger claim about SHA-1 than *high*.

The interesting quantity is never the measurement. It is the measurement divided by what a random function would have given.

1.3.3 *Sigma counts should be divergences*

Comparing a max-statistic against a per-bin uniform gives 775,000× where the sigma-delta null gives 184.5×, the error the distinguishing literature resolves [4]. A divergence is the right currency because it converts directly into the number of samples any distinguisher would need, the only thing deciding whether a structure is exploitable. The conversion is not made here, and it is the largest unresolved methodological debt.

1.4 THE COMPILER AUDIT

A compiler folds, contracts, reassociates and hoists, and a folded measurement is indistinguishable by eye from a computed one. Results depending on the optimizer are not results.

Two arms, both mechanical. **Differential**: build every bench three ways, `-O0`, `-O2`, and `-O2 -ffp-contract=off`, and run all three. A bench whose three outputs are byte-identical computed its numbers instead of inheriting them, and since GCC contracts multiply-add into FMA by default, the third arm separates that from folding proper. **Constant pool**: read every double in a binary's read-only data and report which of the numbers the bench printed were already sitting there before it ran.

Neither arm proves a folded constant was folded *correctly*, a third failure mode, and for the Rényi bench that is closed separately by reimplementing every printed prediction in another language against another libm.

1.5 FOUR PRINCIPLES APPLIED TO THIS CHAPTER

The orior workbook [11] establishes four principles, each on constructed domains where the answer is known by design instead of inferred from the case that suggested it. Three bear directly here and one reframes this entire chapter.

1.5.1 *Histogram quantities are free, and everything above is a histogram quantity*

Histogram quantities are permutation invariant. They describe the max entropy case and are free. The arrangement remains after that.

Every number the Rényi bench produces is a histogram quantity. All seven Rényi orders, the Bellare-Kohno balance, the count of holes, the Arikan guessing factor: each is a function of the multiset of bin counts and of nothing besides. Shuffling which value carries which count leaves every one of them exactly unchanged.

The headline result, that SHA256d's output histogram over a completely enumerated 2^{32} nonce domain is that of a random function to six significant figures, is therefore entailed instead of discovered. It is the max-entropy case and it was always going to be the answer. That is worth having as a measured fact instead of an assumption, and it is not a finding about the arrangement, because no permutation-invariant quantity can be.

1.5.2 *The null must delete the property being asked about*

The principle is established on four constructed clouds, lattice or scatter crossed with patterned or random values, where a value-permuting null can only reveal order in the values and a position-moving null can only reveal order in the geometry. The predicted hits fall on a diagonal, and anything off it refutes the principle.

The permutation null over arrangements of a fixed multiset satisfies this exactly for the question asked here: it preserves every count and destroys every position. It deletes the arrangement and retains the counts. Drawing uniformly from the arrangements of a fixed multiset is the least committal distribution consistent with the observed histogram. It asserts nothing beyond the quantity already measured. Every other background in this work is a model and can be false. This one is the data with one property deleted.

1.5.3 *A quantity that fails to conserve indicates the instrument*

State what a measure must be invariant to and check each, instead of inferring the rule from the case that suggested it. Written out for the deficit measured here:

transformation	required
relabel the window values by any fixed bijection	exactly invariant, the histogram is unchanged
reverse or permute the order of nonce enumeration	exactly invariant, counts do not see order
split the domain and recombine the counts	exactly invariant
read a different window position	invariant within the stated spread
run on a different block header	invariant within the stated spread, **not checked**

The last row is the gap. Every measurement in this work is on one header. One header is one corpus, and the homogeneity check the principle demands is not run. It is the failure mode of a single seed, applied to the header instead of to the seed.

The principle's other half is sharper still: the spread over seeds is the floor below which no difference between two corpora means anything, and that floor is used throughout this work and not measured. Here the ratios 1.00967 for SHA and 0.98900 for splitmix differ by about 2% at the byte windows, against an analytic per-family floor of 1.57%. That comparison is sound only because the floor is derivable exactly for a multinomial. Where a floor is not derivable it has to be measured by reseeding, and no measurement here does that.

1.5.4 *The symbol width has to match the scale of the structure*

A detector is not told where the units begin, and a slice of the right width at the wrong offset splits every unit across two symbols. Both the width and the phase have to be swept.

Every window in the Rényi bench is byte-aligned: 8 bits at 32 positions, 16 at 16, 32 at 2. Swept at eight phases, the sixteen-bit family reads a spread ratio of 0.997 against its control. There is no alignment dependence.

1.5.5 *The content has to be printed, not only the statistic*

In the workbook, nine problems are found by reading output and none by a statistic going out of range. This work prints statistics almost exclusively. The exception is the leading-zero table: extended past 44 bits, it shows that the single deep nonce is the block's own solved nonce, rediscovered by the enumeration. That is the strongest external check available here, and it sits one row below a 44-bit cut.

1.5.6 *The cost term is the result*

Widening the search for structure in the hole arrangement, one step at a time:

what was checked	directions	what it cost	what it found
single-bit flips of the count field	32	one pass over 4 GB, seconds	nothing, max 2.28σ
every mask inside any two bytes	393,210	seconds	nothing, max 5.03σ against a control at 4.82
every mask there is	4,294,967,295	16 GB, 1.4×10^{11} butterflies, minutes	nothing, max 6.79σ against an expected 6.24

Each step multiplies the cost by orders of magnitude and returns the same answer. The free half is free: the output histogram matches a random function's to six significant figures, as it must. The other half is made of checks, and the number of checks scales with the space being checked. Testing every linear direction over 2^{32} values costs $n \log n$; testing every quadratic one costs more; the set of structures that could exist is larger than the space itself by an amount no budget closes.

This is the asymmetry that makes mining work, arriving from the other side. Knowing the function completely removes none of the checks. It answers *we know the hash, we know what they did to it*: knowing the transform does not reduce the count of things that have to be verified, and the count is the cost.

1.5.7 *The knee, and where it sits*

The cost does not merely grow. It floors, with a knee, and the knee's position is derivable.

Checking N directions means taking a max over N statistics. The threshold a real signal has to clear is the expected largest of N nulls. Reach goes as $R(N) = \sqrt{2 \ln N}$ minus a slowly varying correction, and cost as $C(N) \sim N \log N$ in time and N in memory. Inverting and substituting gives $N = e^{R^2/2}$ and $C \sim e^{R^2/2}(R^2/2)$.

The cost is exponential in the square of the reach.

reach wanted	directions needed	relative to what was run
4.57σ	393,210	1/10,900
6.24σ	4.29×10^9	**1, and it cost 16 GB and minutes**
8σ	$\sim 4.3 \times 10^{15}$	$\sim 10^6\times$, about 17 petabytes
10σ	$\sim 7.7 \times 10^{23}$	not on this planet

Going from every two-byte mask to every mask is 10,900 times the work and buys 1.67σ. The next 1.67σ costs about a million times more than that, and the one after costs ten billion times more again. The complete-direction approach floors at about 7σ for any budget that exists.

The knee is behind the measurement already made, and the way past it is not more directions. More directions is the move whose price is exponential in the square of what it buys. Getting further needs a statistic where a signal, if there is one, is concentrated in a few places known in advance instead of spread across billions of equally plausible ones, and that is a statement about having a hypothesis and not about having a bigger machine.

1.6 OPEN QUESTIONS

A second header. One block's header is one corpus, and nothing here is asked to conserve across a different one. The enumeration runs on the device and the check is cheap.

A reduced-width analog. An 8-bit-word SHA-256 has a state small enough to enumerate completely. That replaces every sampled result with an exact one instead of buying another $\sqrt{2 \ln N}$ of reach against an exponential price. It is the only open item that is a better object instead of a bigger search.

The order one half re-read of every histogram collected. It is cheap, and it changes what the existing advantage figures mean.

Divergences in place of sigma counts, throughout.

The functional graph proper. Image size, rho and tail lengths, component structure of the compression function as a map on a restricted domain, against Flajolet-Odlyzko's constants.

Whether the deficit line stays straight at reduced rounds. Round eight is where closeness dies; the order axis is not swept at a partial round, and a curvature that appears and then flattens would locate the mixing in a coordinate nothing here uses.

2

THE ROOM VIEWER, MEASURED BEFORE IT IS SPLIT

This chapter states the case for modularizing the room viewer, the browser page that draws the instrument's result, as counts instead of impressions, and states what has to exist before the split.

2.1 WHAT THE FILE IS

	room	voxel	sphere
lines	4511	2130	774
size	211 KB	86 KB	29 KB
script tags	2	2	2
of those, modules	0	0	0
top-level declarations	355	0	50
top-level functions	102	0	24
duplicate top-level names	0	0	0
banner-marked sections	38	0	16
frame-loop sites	3	4	3

No script carries `type="module"`. Every declaration at the top of the room viewer's script is therefore a property of one hoisted scope, and any of the 355 is reachable from any of the 102 functions. Nothing there is private, and nothing is known to be safe to move.

Two numbers say the split is not a rewrite. The duplicate count is zero, and no name has to be renamed to make modules possible. And 38 banner-marked sections already name the decomposition in comments: the boundaries, the engine arms, the beam and what it scatters from, the operation clock, the boundary reconstruction, the panels, the callouts, the redraw. Those sections are the module list.

Declarations are counted at column zero inside the script body. A pattern allowing indentation counts declarations inside functions, and reads 750 declarations and 144 duplicate names, the duplicates being nested scopes doing their job.

2.2 WHAT HAS TO EXIST BEFORE THE SPLIT

A frame-loop checker, and it is not enough on its own. A split of 355 declarations across 38 sections is likelier to drop a value than to stop the loop. A checker answering *did it die* passes a page whose loop turns over data it no longer has.

A watchdog can manufacture the failure it reports. A watchdog that advances its turn counter from `requestAnimationFrame` and runs its check on `setTimeout` declares a page opened in a hidden tab dead on schedule: a background tab stops the first and not the second, and the page completes one turn. If the report latches, setting a flag that the guarded turn returns on, the loop never runs again, even once the tab comes forward. The result is a frozen viewer with a red banner, frozen by its own watchdog.

The checker therefore obeys four rules. A thrown error is fatal. Slowness reports once and stops nothing. A hidden document is not judged at all, and the check re-arms on `visibilitychange`. A loop reaching two turns withdraws a warning raised while it was hidden.

The precondition is not *a* frame-loop checker. It is one that has been run against a page it should pass, in the conditions a reader will actually open it in, because a split inheriting a faulty watchdog inherits a checker that reports a failure it caused. The quantity reported is not always the quantity intended, and a cheap counting check settles which is in hand.

A live page with inert controls passes every such gate. The hash room builder supplies no `clock` key: the page parses cleanly, the loop is watched and running, the server returns 200, and the viewer's clock controls do nothing.

The second checker is harder than it looks. The obvious form scans the template for every `DATA.<key>` read and fails a build whose payload lacks one. Run over the current builders it reports three, and at least one of the three is correct as written.

builder	template	keys the payload lacks
room	room	`clock`, `sources`
hash room	room	`clock`, `sources`
field	voxel	`noteTitle`

Every read of both room keys is guarded. `DATA.clock` appears as `if (DATA.clock && ...)`, as `DATA.clock ? DATA.clock.hold : 0.62`, as `var CLOCK = DATA.clock || null`, and `DATA.sources` appears once as `Math.min(12, DATA.sources || 2)`. A room built without a clock is a supported mode and the fallbacks are deliberate.

Key presence is therefore neither necessary nor sufficient. Not necessary, because a guarded optional key may be absent by design. Not sufficient, because the failure is not a missing reference: the guards work, the defaults apply, and what ships is a live page with inert controls. A checker built on presence alone flags the room builder, which is correct, and it still needs an argument for why the same absence is a defect in the builder beside it.

The check that catches the real failure is a different one: for each builder and template pair, report which optional features end up inert, letting a build state *this page has no clock* out loud instead of shipping a dead control. It reports and does not refuse, failing only where a feature the builder's own name promises comes out inert.

The three rows above are candidates and not verdicts. They are found by matching payload keys as string literals in the builder source, and a key supplied through a variable would read as missing. The room and hash room builders are confirmed by reading them: both pack exactly `shell`, `core`, `source`, `things` and `settings`.

2.3 WHAT THE SPLIT REMOVES FIRST

`OCT_SIGNS` is a hardcoded two-by-two-by-two sign loop carrying the comment *the arms are fixed*. That comment does not hold. An arm is a weight function over the placement and an arm set is a matrix whose rank the reading carries, and the eight sign octants are one instance of that with rank 8, blind in 248 of 256 directions. The loop asserts they are the only instance.

2.4 VECTORIZING

Vectorizing is not surveyed here, and it follows the split instead of leading it. A vectorized inner loop that drops a term is a picture that still draws, and no checker reads a picture.

2.5 DATA AND CODE AVAILABILITY

The benches, the viewer templates and their builders are in the orior repository, `https://github.com/dstroy0/orior`. The benches are self-contained C programs. The SHA-1 balance values are those published by Bellare and Kohno [5]; every other number here is computed by the benches from inputs they generate or enumerate.

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